

Amio Rashid

Email — Github — LinkedIn — Portfolio

Education

B.Sc. in Computer Science and Engineering

University of Dhaka

Experience

Co-Founder

Rokkhe, Bangladesh (July 2026 – Present)

- Co-founded Rokkhe, an autonomous AI penetration testing platform focused on validating vulnerabilities through demonstrable exploits.

Software Engineering Intern

Ohkuchi Electronics, Kagoshima, Japan (February 2026)

- Completed a one-month internship as a Software Engineering Intern at Ohkuchi Electronics, contributing to industrial automation and AI-based vision system development.
- Collaborating with hardware and system engineering teams to test and validate the solution on real factory devices and field samples.

Bangla AI Trainer & QA

Scale AI, Remote (Jan 2024 – August 2024)

- Generated high-quality Bangla training data for large language models across diverse domains, including prompt-response creation and refinement.
- Reviewed, evaluated, and improved AI-generated responses to support model alignment, quality assurance, and Reinforcement Learning with Human Feedback (RLHF) processes.

Publications

[1] **Trajectory-Informed Multi-Camera Placement for Football Player Tracking: A Particle Swarm Optimization Approach.** *3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026)*, 2026.

Research Experience

Research Assistant

2026 – Present

Green Networking Research Lab (GNR), University of Dhaka

Research Collaborator

2025 – Present

Dept. of CSE, University of Dhaka

Research Focus

I am interested in green networking, edge intelligence and computing, the Internet of Things (IoT), mobile crowdsourcing systems, and network intelligence. My research focuses on developing algorithms, protocols, and intelligent systems for sustainable, energy-efficient, and resource-aware networking and computing.

I am particularly interested in leveraging AI, IoT, and data-driven technologies for carbon emission monitoring, environmental sustainability, and intelligent carbon management solutions.

I am also interested in applying AI and optimization techniques to industrial IoT, digital twins, smart healthcare, and vehicular edge computing to build scalable and sustainable next-generation networked systems.

Technical Skills

Frontend Skills: React, Next.js, JavaScript

Backend Skills: Node.js, Spring Boot, FastAPI, Java, Python, SQL, MySQL, SQLite, PostgreSQL, Oracle

Agentic AI Skills: AI Workflow Automation, LLM Integration, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Chatbot Development, Hugging Face, OpenAI API, Ollama

DevOps Skills: Docker, Git, GitHub, Linux, Nginx

Tools: VSCode, IntelliJ, LaTeX

Application Development Projects

AI-Assisted Student Helper

- Built an AI-driven student support system combining PDF processing, semantic search, chat-based QandA, and flashcard
- **Tech:** FastAPI, Python, MongoDB, Supabase, LLMs, PDF parsing.
- **Link:** github.com/fayek2025/AI-Assisted-Student-Helper

PMICS Administration Management System

- Engineered a production-ready academic and financial administration platform used to streamline student, batch, and institutional operations.
- **Tech:** PostgreSQL, SQL, Next.js, Supabase.
- **Link:** github.com/Flamy-A/FinMan

Cafe Shop Management System

- Led development of a Java-based desktop application to streamline cafe operations including order processing, employee management, billing, inventory, and reporting.
- Implemented object-oriented design principles using JavaFX, FXML, and JDBC for modular and maintainable architecture.
- **Tech:** Java, OOP, JavaFX, JDBC, FXML.
- **Link:** github.com/Flamy-A/Cafe-Management-System

Project Minesweeper with Treasure Hunt

- Developed a feature-rich C++ Minesweeper game using Qt with Easy, Medium, Hard, and Custom modes, including a progressive story mode.
- Implemented randomized bomb placement, grid customization, bomb flagging logic, and integrated timer functionality.
- **Tech:** C++, Qt Creator.
- **Link:** github.com/TanzilaKhan1/Project_minesweeper_with_treasure_hunt

Languages

- Bengali, English
- Japanese – JLPT N5 Level (Basic Conversation Ability)

Honors and Awards

Champion, Capture The Flag – BUP CSE Fest 2026

Sep 2026

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